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The modelling of axisymmetric turbulence

Z.W.T. Mason

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Abstract

The solution space for anisotropic turbulence is bounded on two sides by what is known as axisymmetric turbulence. It is shown that, for such turbulence, a basis needs only two terms - the Kronecker delta and the anisotropy tensor since the square of the latter is a linear combination of the other two. This type of turbulence also has a fixed relationship between the second and third invariants of the anisotropy tensor which can be simplified to derive a constraint equation for one-dimensional shear flows. The subsequent derivation of a model which satisfies this constraint equation is only partially successful but raises multiple issues with the Reynolds stress models shipped with commercial software and with some instances of DNS data.

1. Introduction

Turbulence models for isotropic turbulence are well established. However, the flows of interest are anisotropic which requires more complicated modelling and more expense in the solving due to increased time and storage space thus requiring more accuracy to justify their usage. For these flows there are multiple models whereas the physics only requires one. These models are then tuned in an attempt to minimise the error between the model and the flows of particular interest resulting in model constants that lack an ability to be derived from a combination of the mathematical and physical constants.

Starting from first principles an equation for the transport of the Reynolds stresses may be derived from the Navier-Stokes equations as

$$\frac{\partial \overline{u_i u_j}}{\partial t} + U_k \frac{\partial \overline{u_i u_j}}{\partial x_k} = \left(\phi_{ij2} + P_{ij}\right) + \frac{\partial}{\partial x_k}\left(\nu \frac{\partial \overline{u_i u_j}}{\partial x_k} + D_{ijk}\right) + \left(\phi_{ij1} - \epsilon_{ij}\right) \tag{1}$$

where

$$P_{ij} \equiv -\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k},\ D_{ijk} \equiv -\overline{u_i u_j u_k} - \frac{1}{\rho}\overline{p u_j}\delta_{ik} - \frac{1}{\rho}\overline{p u_i}\delta_{jk},\ \epsilon_{ij} \equiv 2\nu\overline{\frac{\partial u_i}{\partial x_k}\frac{\partial u_j}{\partial x_k}},\ \phi_{ij1} + \phi_{ij2} \equiv \overline{\frac{p}{\rho}\left(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}\right)} \tag{2}$$

A closure of the above equation is sought in the variables U_i, $\overline{u_i u_j}$ and $\epsilon \equiv \epsilon_{ii}/2$ to produce a Reynolds Stress Models (RSM). Existing models are said to be temperamental resulting in low uptake, there are a number of competing closures as shown in table 1. The Imperial College isotropization of production model of Launder, Reece and Rodi [1], its quasi-isotropic variant and potentially utilising the Gibson-Launder wall-reflection [2]. The NASA Langley model of Speziale, Sarkar and Gatski [3]. The Elliptic Blending RSM of Manceau [4]. The stress-omega model of Wilcox [5].

Email address: papers@stochanswers.com (Z.W.T. Mason)

	LRR-IOP	LRR-QI	SSG	EBRSM	SO
OpenFOAM	✓		✓	✓	
FLUENT	✓		✓		✓
CFX	✓	✓	✓		✓
AVL Fire			✓		

Table 1: Popular RSMs

The anisotropy tensor is defined in terms of the Reynolds stresses and their contraction the turbulent kinetic energy $k \equiv \overline{u_i u_i}/2$

$$b_{ij} \equiv \frac{\overline{u_i u_j}}{2k} - \frac{1}{3}\delta_{ij} \tag{3}$$

The invariants of the characteristic equation of which are

$$I_2 \equiv -\frac{1}{2} b_{ij} b_{ji},\ I_3 \equiv det\left(b_{ij}\right) \tag{4}$$

The characteristic equation may be solved by Cardano's formula. The roots are real if the barycentric co-ordinates have specific signs. For axisymmetric turbulence (rod-like for I_3 positive and disc-like for negative)

$$\frac{I_2^3}{27} + \frac{I_3^2}{4} \leq 0 \tag{5}$$

and for two-component turbulence (at the wall)

$$F \equiv 1 + 27 I_3 + 9 I_2 \geq 0 \tag{6}$$

These bounds on the roots form what is known as the Lumley triangle. The first has equality for rod-like and disc-like axisymmetric turbulence. The second has equality for 2 component turbulence such as found at a wall. Examples calculated by Direct Numerical Simulation of turbulence are given in figures 1-6. Figure 1 suggests that the free-stream boundary condition used for the zero-pressure gradient boundary layer is wrong since having achieved rod-like turbulence the numbers jump to disc-like turbulence at the free stream rather than go to isotropy. Figure 6 suggests that a mixing layer results in rod-like turbulence but the particular simulation is noisy leading to some disc-like turbulence.

An orthogonal basis for the Reynolds stresses may be composed from their zeroth to second powers

$$\delta_{ij},\ b_{ij},\ B_{ij} \equiv b_{ik} b_{kj} + \frac{2}{3} I_2 \delta_{ij} + \frac{3 I_3}{2 I_2} b_{ij} \tag{7}$$

For axisymmetric turbulence there are only two independent values in the Reynolds stress tensor and this results in $B_{ij} = 0$.

For a one-dimensional flow with axisymmetric turbulence $\overline{uw} = \overline{vw} = 0$ and the equation

$$\frac{I_2^3}{27} + \frac{I_3^2}{4} = 0 \tag{8}$$

can be expanded into a cubic equation in

$$b_{11} b_{22} - b_{12}^2 \tag{9}$$

with a double root at $-2b_{33}^2$ and a single root at $b_{33}^2/4$. In particular for channel flow in the 1 direction with walls in the 2 direction

$$b_{11} b_{22} - b_{12}^2 + 2 b_{33}^2 = 0 \tag{10}$$

This can be differentiated with respect to time and the time derivatives replaced by one of the production, diffusion and dissipation terms. Differentiating equation 3 with respect to time results in

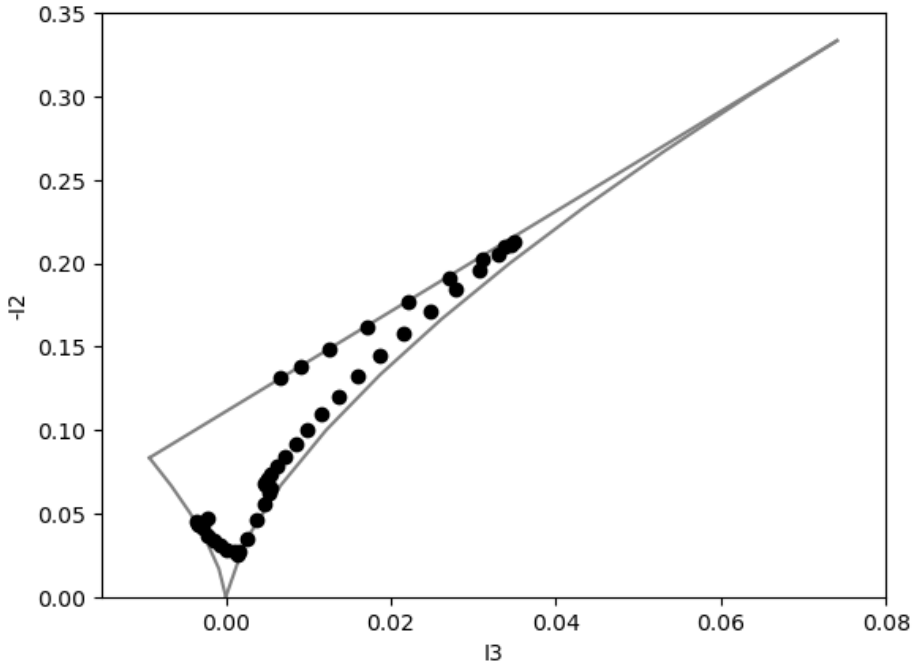

Figure 1: DNS of zero-pressure gradient boundary layer [6]

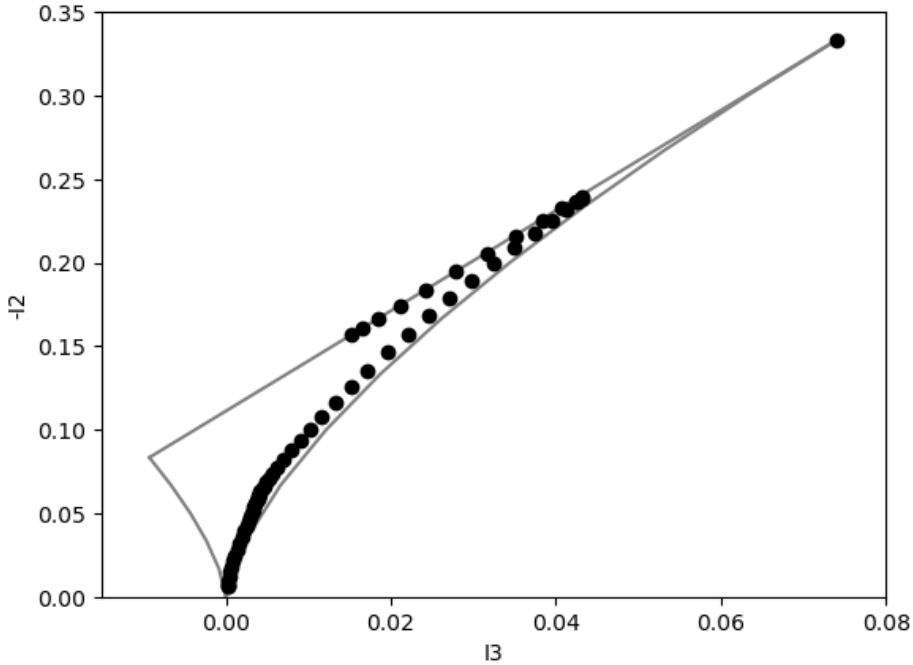

Figure 2: DNS of channel flow [7]

$$b_{11}\frac{\partial b_{22}}{\partial t} + b_{22}\frac{\partial b_{11}}{\partial t} - 2b_{12}\frac{\partial b_{12}}{\partial t} + 4b_{33}\frac{\partial b_{33}}{\partial t} = \frac{1}{2k}\left(b_{11}\frac{\partial \overline{vv}}{\partial t} + b_{22}\frac{\partial \overline{uu}}{\partial t} - 2b_{12}\frac{\partial \overline{uv}}{\partial t} + 4b_{33}\frac{\partial \overline{ww}}{\partial t} - 2b_{33}\frac{\partial k}{\partial t}\right) \quad (11)$$

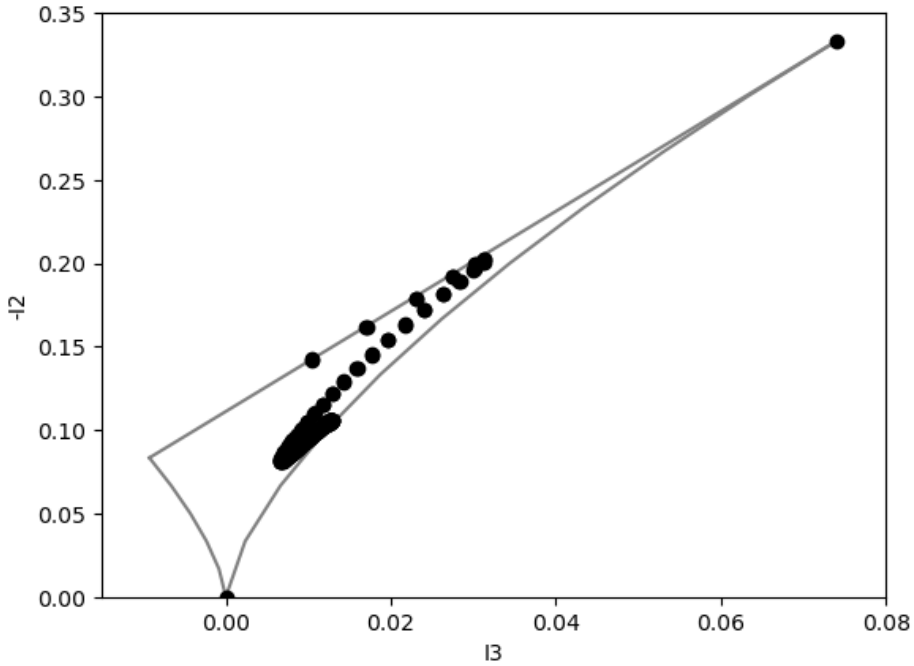

Figure 3: DNS of Couette flow [8]

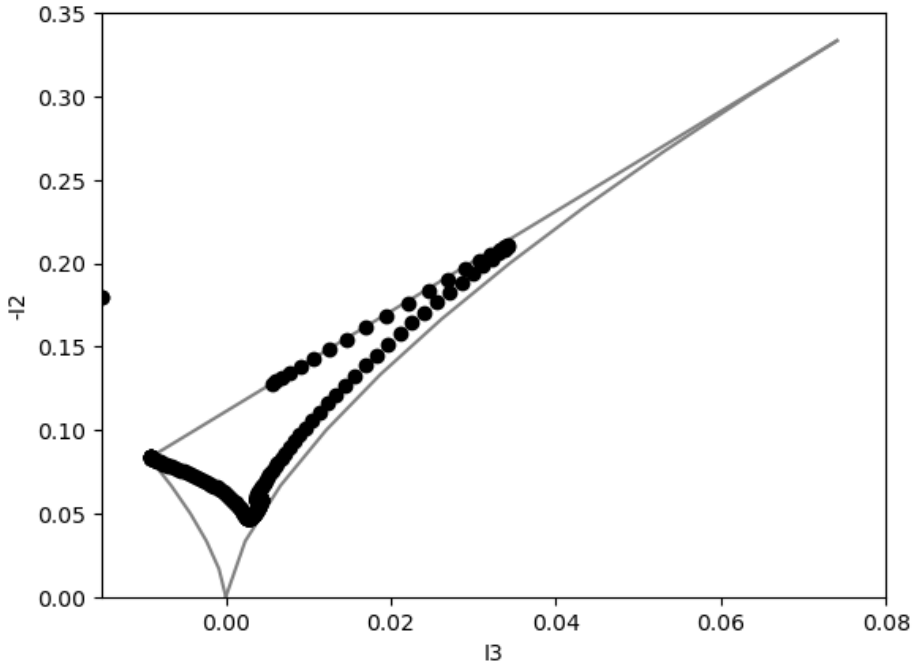

Figure 4: DNS of Couette Poiseuille flow [9]

Expanding the discussion to the general case of full anisotropy a number of flows have one or more components of the shear Reynolds Stresses are identically zero. For this to occur in general requires that, for the specific shear

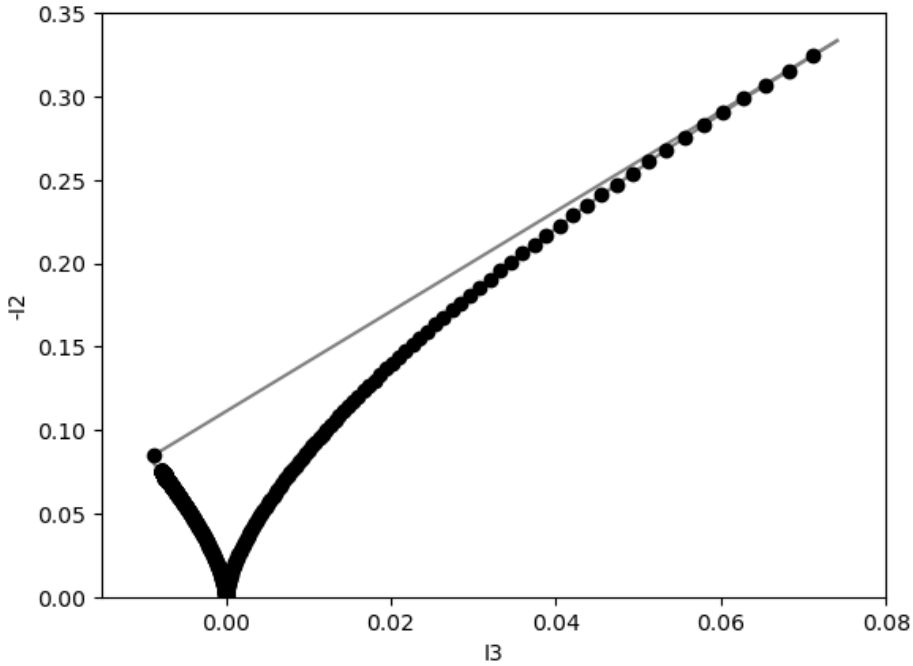

Figure 5: DNS of the decay of turbulence [10]

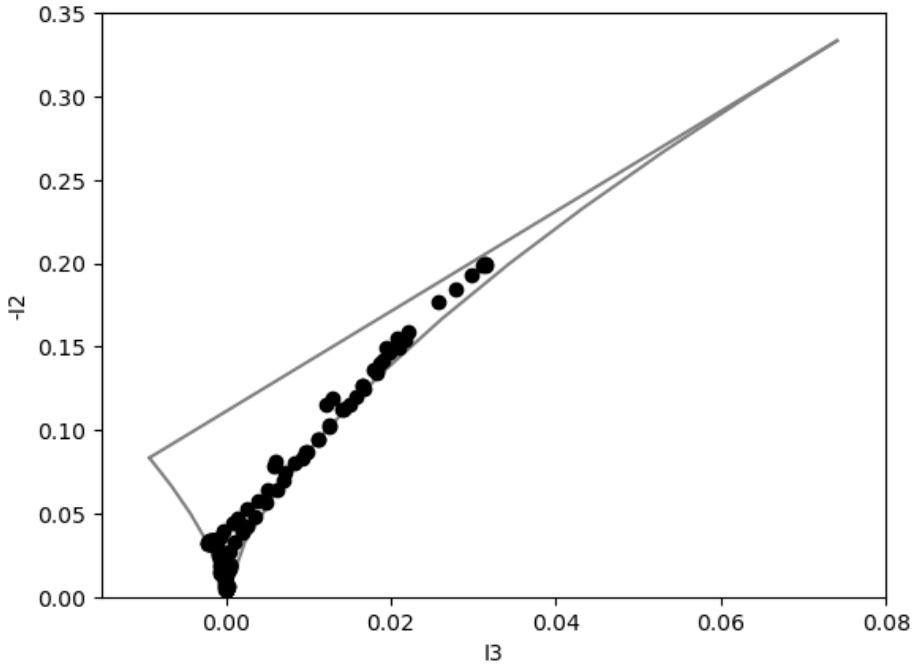

Figure 6: DNS of a mixing layer [11]

component the production, diffusion and dissipation terms are also identically zero.

At the wall ($F = 0$) there is two component turbulence which may be described, in terms of a proportion p, as

$$b_{ij} = \mathrm{diag}\,(p - 1/3, -1/3, 2/3 - p) \tag{12}$$

hence, with a double root ($p = 0, 1$) for one-component turbulence and a single root ($p = 1/2$) for two-component axisymmetric turbulence,

$$-6I_2 B_{ij} = p\,(p-1)\,(2p-1)\,\mathrm{diag}\,(p-1, 1-2p, p) \tag{13}$$

2. Dissipation terms

Expanding the dissipation tensors as a set of orthogonal components

$$\frac{\epsilon_{ij} - \phi_{ij1}}{\epsilon} = \frac{2}{3}\delta_{ij} + C_1 b_{ij} + C_2 B_{ij} \tag{14}$$

Choi and Lumley [12] showed that anisotropic turbulence decays towards axisymmetric turbulence which implies that $C_2 \neq 0$. This is contrary to the model of Rotta [13] which is used by LRR with $C_1 = 3.6$. For SSG $C_1 = 3.4$ and $C_2 = -4.2$.

The dissipation tensor itself was modelled by Hallbäck, Groth and Johansson [14]. Note that $a_{ij} = 2b_{ij}$, $A_2 = a_{ij}a_{ij}$.

$$\frac{\epsilon_{ij}}{\epsilon} = \frac{2}{3}\delta_{ij} + \left(1 + \frac{3}{4}\left(\frac{A_2}{2} - \frac{2}{3}\right)\right)a_{ij} - \frac{3}{4}\left(a_{ik}a_{kj} - \frac{1}{3}A_2\delta_{ij}\right) \tag{15}$$

Substitution of the expansion into equation 11 results in automatic satisfication

$$b_{11}\frac{\partial\overline{vv}}{\partial t} + b_{22}\frac{\partial\overline{uu}}{\partial t} - 2b_{12}\frac{\partial\overline{uv}}{\partial t} + 4b_{33}\frac{\partial\overline{ww}}{\partial t} - 2b_{33}\frac{\partial k}{\partial t} \sim 2C_1\epsilon\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) = 0 \tag{16}$$

The corresponding terms for the anisotropy transport are

$$\frac{1}{2k}\left(\epsilon_{ij} - \phi_{ij1} - \left(b_{ij} + \frac{1}{3}\delta_{ij}\right)2\epsilon\right) = \frac{\epsilon}{2k}\left((C_1 - 2)\,b_{ij} + C_2 B_{ij}\right) \tag{17}$$

hence anisotropy decays if and only if $C_1 > 2$. By power series expansion it can be shown that, seemingly, at the wall $C_1 = 2$ and $C_2 = 0$ due to a balance between dissipation and viscous diffusion.

For the decay of turbulence, the change in anisotropy may be expressed as

$$\frac{\partial b_{ij}}{\partial t} = -\frac{\epsilon}{2k}\left((C_1 - 2)\,b_{ij} + C_2 B_{ij}\right) \tag{18}$$

thus

$$\frac{\partial I_2}{\partial t} = -b_{ij}\frac{\partial b_{ij}}{\partial t} = \frac{\epsilon b_{ij}}{2k}\left((C_1 - 2)\,b_{ij} + C_2 B_{ij}\right) = -\frac{\epsilon}{k}\,(C_1 - 2)\,I_2 \tag{19}$$

and, on the principal axes,

$$\frac{\partial I_3}{\partial t} = \frac{\partial}{\partial t}\,(b_{11}b_{22}b_{33}) = -\frac{\epsilon}{2k}\,(3\,(C_1 - 2)\,I_3 + C_2\,(b_{22}b_{33}B_{11} + b_{11}b_{33}B_{22} + b_{11}b_{22}B_{33})) \tag{20}$$

since

$$b_{22}b_{33} + b_{11}b_{33} + b_{11}b_{22} = I_2 \tag{21}$$

then

$$\frac{\partial I_3}{\partial t} = -\frac{\epsilon}{2k}\left(3\,(C_1 - 2)\,I_3 + C_2\left(\frac{2}{3}I_2^2 + \frac{9I_3^2}{2I_2}\right)\right) \tag{22}$$

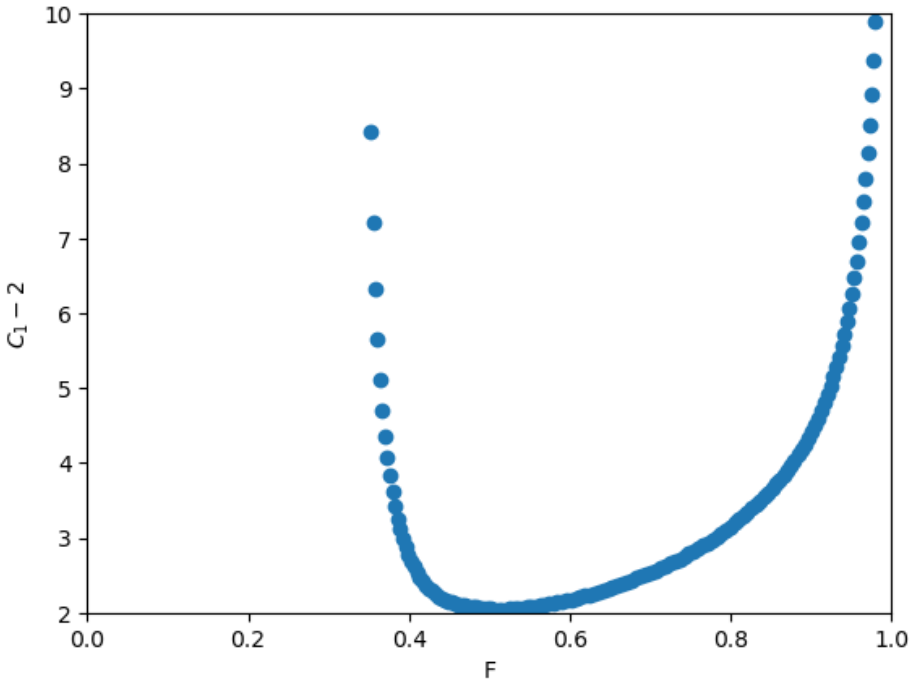

Figure 7: DNS of the decay of turbulence [10]

Expanding the Reynolds stress equations in terms of distance from the wall shows laminar diffusion and dissipation balance at the wall

$$\nu\frac{\partial^2\overline{u_iu_j}}{\partial y^2} \sim 2\nu\frac{\overline{u_iu_j}}{y^2} = 4\nu k\frac{b_{ij}+\delta_{ij}/3}{y^2} \sim$$
$$\epsilon\left(\frac{2}{3}\delta_{ij}+C_1b_{ij}+C_2B_{ij}\right) \sim 2\nu\frac{k}{y^2}\left(\frac{2}{3}\delta_{ij}+C_1b_{ij}+C_2B_{ij}\right),\ ij = 11 \text{ or } 33$$
$$\Rightarrow (C_1-2)\,b_{ij}+C_2B_{ij} \sim 0,\ ij = 11 \text{ or } 33 \quad (23)$$

Since $C_1 > 2$ it is required for B_{11} to be proportional to b_{11} and that B_{33} is in the same proportion to b_{33}. Thus, equations 12 and 13 imply

$$p-1 = \lambda\,(p-1/3)\,,\ p = \lambda\,(2/3-p) \quad (24)$$

Eliminating λ results in a quadratic equation with complex roots

$$p = \frac{1}{2} \pm i\sqrt{\frac{1}{12}} \quad (25)$$

Thus, there is an imbalance at the wall which seems to imply that the dissipation terms cannot be modelled using merely the powers of the anisotropy tensor.

The DNS of the decay of both rod and disc like turbulence was undertaken by Livescu and Ristorcelli [10]. The resulting plot for equation 19 is given in figure 7. Obviously this flow is not a stationary process and, in the absence of experimental data to compare against, the reader needs to be sceptical of the resulting curve.

At the centreline in channel flow there is no production thus

$$\frac{\phi_{ij}-\epsilon_{ij}}{\epsilon_{kk}} + \frac{1}{3}\delta_{ij} = -C_1b_{ij} \quad (26)$$

The results for calculating C_1 from DNS results are given in table 2. Note that if the simulation had resulted in exactly axisymmetric turbulence rather than approximate then the 11 and 22 components would be equal. For the values being roughly equal they are greater than 2 and thus the anisotropy would decay according to equation 19.

Source	Re	11	22	33
Kim	180	2.187351081	1.614136623	2.802851851
Kim	395	3.28781994	2.367964882	4.422307844
Kasagi	150	2.208065551	2.175915602	2.244168959
Horiuti	180	3.615033977	2.60232752	4.745236763
Horiuti	180	3.615033977	2.60232752	4.745236763
AJZM	180	2.40086994	2.185085768	2.619202744
AJZM	550	2.388149973	2.152474258	2.610285605
AJZM	950	2.391965281	2.254800036	2.523968168
AJZM	2000	2.454661779	2.271890084	2.655244035

Table 2: Values for C_1 from channel flows

3. Production terms

The LRR-IOP (Isotropisation of Production) model is

$$P_{ij} + \phi_{ij2} = -\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k} - C_2\left(-\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k} - \frac{1}{3}P_{kk}\delta_{ij}\right) \tag{27}$$

alternatively written as

$$P_{ij} + \phi_{ij2} = (C_2 - 1)\,\frac{2}{3}k\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) + \frac{1}{3}P_{kk}\delta_{ij} + (C_2 - 1)\,2k\left(b_{jk}\frac{\partial U_i}{\partial x_k} + b_{ik}\frac{\partial U_j}{\partial x_k} - \frac{2}{3}b_{lk}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right) \tag{28}$$

where $C_2 = 0.6$
The LRR-QI (Quasi-Isotropic) model is

$$\begin{aligned} P_{ij} + \phi_{ij2} = {} & -\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k} - \frac{C_2 + 8}{11}\left(-\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k} - \frac{1}{3}P_{kk}\delta_{ij}\right) \\ & - \frac{30C_2 - 2}{55}k\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) - \frac{8C_2 - 2}{11}\left(-\overline{u_i u_k}\frac{\partial U_k}{\partial x_j} - \overline{u_j u_k}\frac{\partial U_k}{\partial x_i} - \frac{1}{3}P_{kk}\delta_{ij}\right) \end{aligned} \tag{29}$$

alternatively written as

$$\begin{aligned} P_{ij} + \phi_{ij2} = {} & -\frac{17C_2 + 15}{11}\frac{2}{3}k\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) + \frac{1}{3}P_{kk}\delta_{ij} + \frac{C_2 - 3}{11}2k\left(b_{jk}\frac{\partial U_i}{\partial x_k} + b_{ik}\frac{\partial U_j}{\partial x_k} - \frac{2}{3}b_{lk}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right) \\ & + \frac{8C_2 - 2}{11}2k\left(b_{ik}\frac{\partial U_k}{\partial x_j} + b_{jk}\frac{\partial U_k}{\partial x_i} - \frac{2}{3}b_{lk}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right) \end{aligned} \tag{30}$$

The SSG model is

$$\begin{aligned} P_{ij} + \phi_{ij2} = {} & -\overline{u_j u_k}\frac{\partial U_i}{\partial x_k} - \overline{u_i u_k}\frac{\partial U_j}{\partial x_k} - \frac{1}{2}C_{r1}P_{kk}a_{ij} + C_{r2}kS_{ij} - C_{r3}kS_{ij}\sqrt{a_{mn}a_{mn}} \\ & + C_{r4}k\left(a_{ik}S_{jk} + a_{jk}S_{ik} - \frac{2}{3}a_{kl}S_{kl}\delta_{ij}\right) + C_{r5}k\left(a_{ik}\Omega_{jk} + a_{jk}\Omega_{ik}\right) \end{aligned} \tag{31}$$

where $C_{r1} = 0.9$, $C_{r2} = 0.8$, $C_{r3} = 0.65$, $C_{r4} = 0.625$, $C_{r5} = 0.2$ and

$$S_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{32}$$

$$\Omega_{ij} = \frac{1}{2}\left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i}\right) \tag{33}$$

alternatively written as

$$\begin{aligned} P_{ij} + \phi_{ij2} = &\left(-\frac{4}{3} + C_{r2} - 2C_{r3}\sqrt{-2I_2}\right)kS_{ij} - 2kb_{jk}\frac{\partial U_i}{\partial x_k} - 2kb_{ik}\frac{\partial U_j}{\partial x_k} - C_{r1}P_{kk}b_{ij} \\ &+ C_{r4}k\left(b_{ik}\frac{\partial U_j}{\partial x_k} + b_{jk}\frac{\partial U_i}{\partial x_k} - \frac{2}{3}b_{kl}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right) + C_{r5}k\left(b_{ik}\frac{\partial U_j}{\partial x_k} + b_{jk}\frac{\partial U_i}{\partial x_k}\right) \\ &+ C_{r4}k\left(b_{ik}\frac{\partial U_k}{\partial x_j} + b_{jk}\frac{\partial U_k}{\partial x_i} - \frac{2}{3}b_{kl}\frac{\partial U_k}{\partial x_l}\delta_{ij}\right) - C_{r5}k\left(b_{ik}\frac{\partial U_k}{\partial x_j} + b_{jk}\frac{\partial U_k}{\partial x_i}\right) \end{aligned} \tag{34}$$

Expanding the production tensors in the set of orthogonal components

$$P_{ij}+\phi_{ij2} = C_1k\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)+X_{ij}P_{kk}+Y_{lk}k\left(\delta_{lj}\frac{\partial U_i}{\partial x_k} + \delta_{li}\frac{\partial U_j}{\partial x_k} - \frac{2}{3}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right)+Z_{lk}k\left(\delta_{lj}\frac{\partial U_k}{\partial x_i} + \delta_{li}\frac{\partial U_k}{\partial x_j} - \frac{2}{3}\frac{\partial U_k}{\partial x_l}\delta_{ij}\right) \tag{35}$$

where

$$X_{ij} = \frac{1}{3}\delta_{ij} + C_2b_{ij} + C_3B_{ij} \tag{36}$$

$$Y_{lk} = C_4b_{lk} + C_5B_{lk} \tag{37}$$

$$Z_{lk} = C_6b_{lk} + C_7B_{lk} \tag{38}$$

$$\overline{u_iu_j} = 2k\left(b_{ij} + \frac{1}{3}\delta_{ij}\right) \Rightarrow P_{kk} = -2\overline{u_lu_k}\frac{\partial U_l}{\partial x_k} = -4kb_{lk}\frac{\partial U_l}{\partial x_k} \tag{39}$$

The coefficients of the described models are given in table 3.

	C1	C2	C4	C6	C3, C5, C7
LRR-IOP	-4/15	0	-4/5	0	0
LRR-QI	-84/55	0	-24/55	28/55	0
SSG	-8/15 -13/10 sqrt(-2 I2)	-17/30	-47/40	17/40	0

Table 3: Popular RSM coefficients

The corresponding terms for the anisotropy transport are

$$\frac{C_1}{2}\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right)+\left((C_2-1)\,b_{ij} + C_3B_{ij}\right)\frac{P_{kk}}{2k}+\frac{Y_{lk}}{2}\left(\delta_{lj}\frac{\partial U_i}{\partial x_k} + \delta_{li}\frac{\partial U_j}{\partial x_k} - \frac{2}{3}\frac{\partial U_l}{\partial x_k}\delta_{ij}\right)+\frac{Z_{lk}}{2}\left(\delta_{lj}\frac{\partial U_k}{\partial x_i} + \delta_{li}\frac{\partial U_k}{\partial x_j} - \frac{2}{3}\frac{\partial U_k}{\partial x_l}\delta_{ij}\right) \tag{40}$$

In one-dimensional flows where $U = (U_1(x_2), 0, 0)$

$$P_{23} + \phi_{232} = (C_2b_{23} + C_3B_{23})\,P_{kk} + (C_6b_{31} + C_7B_{31})\,k\frac{\partial U}{\partial y} \tag{41}$$

and

$$\overline{uw} = \overline{vw} = 0 \Rightarrow b_{31} = b_{23} = 0 \Rightarrow B_{31} = B_{23} = 0 \Rightarrow P_{23} = 0 \tag{42}$$

$$(P_{11} + \phi_{112})/k\frac{\partial U}{\partial y} = -4b_{12}\left(\frac{1}{3} + C_2 b_{11} + C_3 B_{11}\right) + \frac{4}{3}(C_4 b_{12} + C_5 B_{12}) - \frac{2}{3}(C_6 b_{21} + C_7 B_{21}) \tag{43}$$

$$(P_{22} + \phi_{222})/k\frac{\partial U}{\partial y} = -4b_{12}\left(\frac{1}{3} + C_2 b_{22} + C_3 B_{22}\right) - \frac{2}{3}(C_4 b_{12} + C_5 B_{12}) + \frac{4}{3}(C_6 b_{21} + C_7 B_{21}) \tag{44}$$

$$(P_{33} + \phi_{332})/k\frac{\partial U}{\partial y} = -4b_{12}\left(\frac{1}{3} + C_2 b_{33} + C_3 B_{33}\right) - \frac{2}{3}(C_4 b_{12} + C_5 B_{12}) - \frac{2}{3}(C_6 b_{21} + C_7 B_{21}) \tag{45}$$

$$(P_{12} + \phi_{122})/k\frac{\partial U}{\partial y} = C_1 - 4b_{12}(C_2 b_{12} + C_3 B_{12}) + C_4 b_{22} + C_5 B_{22} + C_6 b_{11} + C_7 B_{11} \tag{46}$$

The corresponding terms for the anisotropy transport are

$$\frac{1}{2k}\left(P_{ij} + \phi_{ij2} + 4kb_{lk}\frac{\partial U_l}{\partial x_k}\left(b_{ij} + \frac{1}{3}\delta_{ij}\right)\right) \tag{47}$$

Substitution into equation 11 (note that $B_{ij} \equiv 0$) results in

$$b_{11}\frac{\partial \overline{vv}}{\partial t} + b_{22}\frac{\partial \overline{uu}}{\partial t} - 2b_{12}\frac{\partial \overline{uv}}{\partial t} + 4b_{33}\frac{\partial \overline{ww}}{\partial t} - 2b_{33}\frac{\partial k}{\partial t} \sim -2C_1 b_{12} - 8(C_2 - 1)b_{12}\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) + \frac{1}{3}C_4 b_{12}(4b_{22} - 2b_{11} - 6b_{22} - 8b_{33}) + \frac{1}{3}C_6 b_{12}(4b_{11} - 2b_{22} - 6b_{11} - 8b_{33}) \tag{48}$$

For the above terms to evaluate to zero requires $C_1 = 0$ and $C_6 = -C_4$.

4. Diffusion terms

The diffusion tensor may be written as

$$D_{ijk} = -\overline{u_i u_j u_k} - \frac{1}{\rho}\left(\delta_{ik}\overline{u_j p} + \delta_{jk}\overline{u_i p}\right) \tag{49}$$

The generalized gradient diffusion model of Daly and Harlow [15] is

$$\epsilon D_{ijk} = C_s k\overline{u_k u_l}\frac{\partial \overline{u_i u_j}}{\partial x_l} \tag{50}$$

where $C_s = 0.22$. Although, due to stability issues, a simpler form is usually preferred

$$\epsilon D_{ijk} = \frac{2}{3}C_s k^2\frac{\partial \overline{u_i u_j}}{\partial x_k} \tag{51}$$

Contracting the above to form the k equation results in the multiplying constant being $C_\mu = 2C_s/3 = 0.14666 \neq 0.09$. N.B. $1/2\pi = 0.159$.

The triple velocity correlation is modelled as

$$-\overline{u_i u_j u_k} = \frac{k^2}{\epsilon}\left(A_{ijk} + A_{jki} + A_{kij}\right) \tag{52}$$

where A_{ijk} is formed from the gradient of the product of three tensors - a mix of anisotropy tensors and Kronecker deltas. This results in the sum of ten terms (fifteen if the gradients of the turbulent kinetic energy are duplicated with gradients of the second invariant) which satisfy symmetry and the consistency equation 11 - the subscripts on the scalars refer to the appendix that the terms are investigated in. Note that the C_{30} term satisfies the conditions when axis-aligned but may not in general.

$$A_{ijk} = C_2 k b_{ij} b_{mn} \frac{\partial b_{mn}}{\partial x_k} + C_{12} k \delta_{ij} b_{km} \frac{\partial b_{mn}}{\partial x_n} + C_{13} k \delta_{ij} b_{mn} \frac{\partial b_{mn}}{\partial x_k} + C_{14} k \delta_{ij} b_{mn} \frac{\partial b_{kn}}{\partial x_m} + C_{27} \delta_{jk} b_{im} b_{mn} \frac{\partial k}{\partial x_n} + C_{28} \delta_{jk} b_{mn} b_{mn} \frac{\partial k}{\partial x_i} + C_{29} \delta_{ij} \frac{\partial b_{kn}}{\partial x_n} + C_{30} k \frac{\partial b_{ij}}{\partial x_k} + C_{31} \delta_{ij} b_{km} \frac{\partial k}{\partial x_m} + C_{32} b_{ji} \frac{\partial k}{\partial x_k} + C_{33} \delta_{ij} \frac{\partial k}{\partial x_k} \quad (53)$$

and the corresponding pressure-velocity correlation is

$$\frac{\epsilon}{k^2} \frac{1}{\rho} \overline{u_i p} = C_2 k b_{ij} b_{mn} \frac{\partial b_{mn}}{\partial x_j} + C_{12} k b_{im} \frac{\partial b_{mn}}{\partial x_n} + C_{13} k b_{mn} \frac{\partial b_{mn}}{\partial x_i} + C_{14} k b_{mn} \frac{\partial b_{in}}{\partial x_m} + C_{27} b_{im} b_{mn} \frac{\partial k}{\partial x_n} + C_{28} b_{mn} b_{mn} \frac{\partial k}{\partial x_i} + C_{29} \frac{\partial b_{in}}{\partial x_n} + C_{30} k \frac{\partial b_{ij}}{\partial x_j} + C_{31} b_{ij} \frac{\partial k}{\partial x_j} + C_{32} b_{ij} \frac{\partial k}{\partial x_j} + C_{33} \frac{\partial k}{\partial x_i} \quad (54)$$

For one-dimensional flows the diffusion of the shear stress reduces to

$$D_{122} = b_{12} \frac{k^2}{\epsilon} \left(C_2 k b_{mn} \frac{db_{mn}}{dy} + C_{32} \frac{dk}{dy} \right) + C_{30} \frac{k^3}{\epsilon} \frac{db_{12}}{dy} + 0 \times (C_{12} + C_{13} + C_{14} + C_{27} + C_{28} + C_{29} + C_{31} + C_{33}) \quad (55)$$

which reduces to just the C_{30} term at the centreline in channel flow and suggests a simpler model for axisymmetric turbulence of

$$A_{ijk} = C_1 k \frac{\partial b_{ij}}{\partial x_k} + C_2 \delta_{ij} b_{km} \frac{\partial k}{\partial x_m} + C_3 b_{ji} \frac{\partial k}{\partial x_k} + C_4 \delta_{ij} \frac{\partial k}{\partial x_k} + C_5 k \delta_{ij} b_{km} \frac{\partial I_2}{\partial x_m} + C_6 k b_{ji} \frac{\partial I_2}{\partial x_k} + C_7 k \delta_{ij} \frac{\partial I_2}{\partial x_k} \quad (56)$$

and

$$\frac{\epsilon}{k^2} \frac{1}{\rho} \overline{u_i p} = C_1 k \frac{\partial b_{ij}}{\partial x_j} + C_2 b_{ij} \frac{\partial k}{\partial x_j} + C_3 b_{ij} \frac{\partial k}{\partial x_j} + C_4 \frac{\partial k}{\partial x_i} + C_5 k b_{ij} \frac{\partial I_2}{\partial x_j} + C_6 k b_{ij} \frac{\partial I_2}{\partial x_j} + C_7 k \frac{\partial I_2}{\partial x_i} \quad (57)$$

where $C_1 - C_7$ are either zeroes, constants or scalar functions of I_2. Note that the forms of the pressure-velocity correlations are not necessarily unique. i.e. C_2 and C_3 multiply the same term as do C_5 and C_6.

This model gives the diffusion of the turbulent kinetic energy as

$$\frac{\epsilon}{k^2} D_{iik} = 3 C_2 b_{ki} \frac{\partial k}{\partial x_i} + 3 C_4 \frac{\partial k}{\partial x_k} + 3 C_5 k b_{ki} \frac{\partial I_2}{\partial x_i} + 3 C_7 k \frac{\partial I_2}{\partial x_k} \quad (58)$$

5. Discussion

This paper has not extended turbulence modelling from the isotropic case to the axisymmetric turbulence subset of anisotropic turbulence. However, it is contended, that a number of useful observations have been made. In particular that the published Reynolds Stress models fail to simulate axisymmetric turbulence since they fail to match the constraints derived herein.

There would appear to be a dearth of high quality Direct Numerical Simulation studies that have had their balance equation published. Such studies should be compared to actual experimental data especially when they simulate a non-stationary process. Should the results vary in only one dimension then a quick sanity check can be made by plotting the invariants of the anisotropy tensor in a Lumley triangle to look at the path made. Note that the wall functions typically used for the Reynolds stress models have constant ratios of said stresses and thus result in a fixed point in the triangle thus failing this sanity check.

The algebraic modelling of the dissipation tensor would appear to be sufficient for axisymmetric turbulence. However, extending to the general case requires transport equations for the full dissipation tensor to ensure, for example, the correct behaviour at a wall. A first pass for such a model could be derived for the axisymmetric turbulence case by differentiating the algebraic model.

Assuming that an algebraic model of the production terms is sufficient then it requires 5 unknown scalars for the case of general anisotropy and only 2 for axisymmetric turbulence.

It is unclear whether the turbulent diffusion can be modelled at the centreline of channel flow. The alternative requires resorting to transport equations for the triple velocity and / or the pressure velocity correlations. The derived model requires 7 scalar functions which will be hard to acquire by fitting of DNS data. It may be possible to integrate Poisson's equation for the pressure fluctuations to acquire a formula for the pressure-velocity correlation and then construct the corresponding triple-velocity correlation.

In conclusion modelling axisymmetric turbulence allows the use of some Mathematical tools which aren't available for the more general case making it somewhat easier.

Appendix A. Diffusion tensors

Appendix A.1. ijkmnmn

$$-\overline{u_i u_j u_k} = C_1 \frac{k^3}{\epsilon} \left(b_{ij} b_{km} \frac{\partial b_{mn}}{\partial x_n} + b_{jk} b_{im} \frac{\partial b_{mn}}{\partial x_n} + b_{ki} b_{jm} \frac{\partial b_{mn}}{\partial x_n} \right) \tag{A.1}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_1 \frac{k^3}{\epsilon} \left(2b_{12}b_{2m} + b_{22}b_{1m}\right) \frac{db_{m2}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.2}$$

At the centreline of channel flow

$$D_{122}(0) = C_1 \frac{k^3}{\epsilon} b_{22} b_{11} \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.3}$$

$$D_{112} = -\overline{uuv} = C_1 \frac{k^3}{\epsilon} \left(b_{11}b_{2m} + 2b_{12}b_{1m}\right) \frac{db_{m2}}{dy} \tag{A.4}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_1 \frac{k^3}{\epsilon} b_{22} b_{2m} \frac{db_{m2}}{dy} + 2\left(C_1 \frac{k^3}{\epsilon} b_{22} b_{2m} \frac{db_{m2}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.5}$$

$$D_{332} = -\overline{wwv} = C_1 \frac{k^3}{\epsilon} b_{33} b_{2m} \frac{db_{m2}}{dy} \tag{A.6}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.7}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \tag{A.8}$$

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$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \tag{A.9}$$

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Appendix A.2. ijmnkmn

$$-\overline{u_i u_j u_k} = C_2 \frac{k^3}{\epsilon} \left(b_{ij} b_{mn} \frac{\partial b_{mn}}{\partial x_k} + b_{jk} b_{mn} \frac{\partial b_{mn}}{\partial x_i} + b_{ki} b_{mn} \frac{\partial b_{mn}}{\partial x_j} \right) \tag{A.10}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_2 \frac{k^3}{\epsilon} b_{12} b_{mn} \frac{db_{mn}}{dy} + \left(C_2 \frac{k^3}{\epsilon} b_{12} b_{mn} \frac{db_{mn}}{dy} - \frac{1}{\rho}\overline{up} \right) \tag{A.11}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.12}$$

$$D_{112} = -\overline{uuv} = C_2 \frac{k^3}{\epsilon} b_{11} b_{mn} \frac{db_{mn}}{dy} \tag{A.13}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_2 \frac{k^3}{\epsilon} b_{22} b_{mn} \frac{db_{mn}}{dy} + 2 \left(C_2 \frac{k^3}{\epsilon} b_{22} b_{mn} \frac{db_{mn}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.14}$$

$$D_{332} = -\overline{wwv} = C_2 \frac{k^3}{\epsilon} b_{33} b_{mn} \frac{db_{mn}}{dy} \tag{A.15}$$

$$\frac{1}{\rho}\overline{u_i p} = C_2 \frac{k^3}{\epsilon} b_{ij} b_{mn} \frac{\partial b_{mn}}{\partial x_j} \tag{A.16}$$

$$\begin{aligned} &b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \\ &C_2 \frac{k^3}{\epsilon} b_{mn} \frac{db_{mn}}{dy} \left(b_{11} b_{22} + b_{22} b_{11} - 2 b_{12} b_{12} + 4 b_{33} b_{33} - b_{33} \times 0 \right) = 2 C_2 \frac{k^3}{\epsilon} b_{mn} \frac{db_{mn}}{dy} \left(b_{11} b_{22} - b_{12}^2 + 2 b_{33}^2 \right) = 0 \end{aligned} \tag{A.17}$$

$$\begin{aligned} &D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ &C_2 \frac{k^3}{\epsilon} b_{mn} \frac{db_{mn}}{dy} \left(b_{22} \frac{db_{11}}{dy} + b_{11} \frac{db_{22}}{dy} - 2 b_{12} \frac{db_{12}}{dy} + 4 b_{33} \frac{db_{33}}{dy} - 0 \times \frac{db_{33}}{dy} \right) = C_2 \frac{k^3}{\epsilon} b_{mn} \frac{db_{mn}}{dy} \frac{d}{dy} \left(b_{11} b_{22} - b_{12}^2 + 2 b_{33}^2 \right) = 0 \end{aligned} \tag{A.18}$$

Appendix A.3. ijmnmkn

$$-\overline{u_i u_j u_k} = C_3 \frac{k^3}{\epsilon} \left(b_{ij} b_{mn} \frac{\partial b_{kn}}{\partial x_m} + b_{jk} b_{mn} \frac{\partial b_{in}}{\partial x_m} + b_{ki} b_{mn} \frac{\partial b_{jn}}{\partial x_m} \right) \tag{A.19}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_3 \frac{k^3}{\epsilon} \left(2b_{12} b_{2n} \frac{db_{2n}}{dy} + b_{22} b_{2n} \frac{db_{1n}}{dy} \right) + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.20}$$

At the centreline of channel flow

$$D_{122}(0) = C_3 \frac{k^3}{\epsilon} b_{22} b_{22} \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.21}$$

$$D_{112} = -\overline{uuv} = C_3 \frac{k^3}{\epsilon} \left(b_{11} b_{2n} \frac{db_{2n}}{dy} + 2b_{12} b_{2n} \frac{db_{1n}}{dy} \right) \tag{A.22}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_3 \frac{k^3}{\epsilon} b_{22} b_{2n} \frac{db_{2n}}{dy} + 2\left(C_3 \frac{k^3}{\epsilon} b_{22} b_{2n} \frac{db_{2n}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.23}$$

$$D_{332} = -\overline{wwv} = C_3 \frac{k^3}{\epsilon} b_{33} b_{2n} \frac{db_{2n}}{dy} \tag{A.24}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.25}$$

$$b_{11} D_{222} + b_{22} D_{112} - 2b_{12} D_{122} + 4b_{33} D_{332} - b_{33} D_{ii2} = \tag{A.26}$$

$$D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2D_{122} \frac{db_{12}}{dy} + 4D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \tag{A.27}$$

Appendix A.4. imjmnkn

$$-\overline{u_i u_j u_k} = C_4 \frac{k^3}{\epsilon} \left(b_{im} b_{jm} \frac{\partial b_{kn}}{\partial x_n} + b_{jm} b_{km} \frac{\partial b_{in}}{\partial x_n} + b_{km} b_{im} \frac{\partial b_{jn}}{\partial x_n} \right) \tag{A.28}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_4 \frac{k^3}{\epsilon} \left(2b_{1m} b_{2m} \frac{db_{22}}{dy} + b_{2m} b_{2m} \frac{db_{12}}{dy} \right) + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.29}$$

At the centreline of channel flow

$$D_{122}(0) = C_4 \frac{k^3}{\epsilon} b_{22} b_{22} \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.30}$$

$$D_{112} = -\overline{uuv} = C_4 \frac{k^3}{\epsilon} \left(b_{1m} b_{1m} \frac{db_{22}}{dy} + 2b_{1m} b_{2m} \frac{db_{12}}{dy} + \right) \tag{A.31}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_4 \frac{k^3}{\epsilon} b_{2m} b_{2m} \frac{db_{22}}{dy} + 2 \left(C_4 \frac{k^3}{\epsilon} b_{2m} b_{2m} \frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.32}$$

$$D_{332} = -\overline{wwv} = C_4 \frac{k^3}{\epsilon} b_{33} b_{33} \frac{db_{22}}{dy} \tag{A.33}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.34}$$

$$b_{11} D_{222} + b_{22} D_{112} - 2b_{12} D_{122} + 4b_{33} D_{332} - b_{33} D_{ii2} = \tag{A.35}$$

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$$D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2D_{122} \frac{db_{12}}{dy} + 4D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \tag{A.36}$$

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Appendix A.5. imjnkmn

$$-\overline{u_i u_j u_k} = C_5 \frac{k^3}{\epsilon} \left(b_{im} b_{jn} \frac{\partial b_{mn}}{\partial x_k} + b_{jm} b_{kn} \frac{\partial b_{mn}}{\partial x_i} + b_{km} b_{in} \frac{\partial b_{mn}}{\partial x_j} \right) \tag{A.37}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_5 \frac{k^3}{\epsilon} b_{1m} b_{2n} \frac{db_{mn}}{dy} + \left(C_5 \frac{k^3}{\epsilon} b_{1m} b_{2n} \frac{db_{mn}}{dy} - \frac{1}{\rho}\overline{up} \right) \tag{A.38}$$

At the centreline of channel flow

$$D_{122}(0) = C_5 \frac{k^3}{\epsilon} b_{11} b_{22} \frac{db_{12}}{dy} + \left(C_5 \frac{k^3}{\epsilon} b_{11} b_{22} \frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up} \right) \tag{A.39}$$

$$D_{112} = -\overline{uuv} = C_5 \frac{k^3}{\epsilon} b_{1m} b_{1n} \frac{db_{mn}}{dy} \tag{A.40}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_5 \frac{k^3}{\epsilon} b_{2m} b_{2n} \frac{db_{mn}}{dy} + 2\left(C_5 \frac{k^3}{\epsilon} b_{2m} b_{2n} \frac{db_{mn}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.41}$$

$$D_{332} = -\overline{wwv} = C_5 \frac{k^3}{\epsilon} b_{33} b_{33} \frac{db_{33}}{dy} \tag{A.42}$$

$$\frac{1}{\rho}\overline{u_i p} = C_5 \frac{k^3}{\epsilon} b_{im} b_{jn} \frac{\partial b_{mn}}{\partial x_j} \tag{A.43}$$

$$\begin{aligned} b_{11} D_{222} + b_{22} D_{112} - 2b_{12} D_{122} + 4b_{33} D_{332} - b_{33} D_{ii2} = \\ C_5 \frac{k^3}{\epsilon} \frac{db_{mn}}{dy} \left(b_{11} b_{2m} b_{2n} + b_{22} b_{1m} b_{1n} - 2b_{12} b_{1m} b_{2n} + 4b_{33} b_{3m} b_{3n} - b_{33} b_{im} b_{in} \right) = \end{aligned} \tag{A.44}$$

$$\begin{aligned} D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2D_{122} \frac{db_{12}}{dy} + 4D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ C_5 \frac{k^3}{\epsilon} \frac{db_{mn}}{dy} \left(b_{2m} b_{2n} \frac{db_{11}}{dy} + b_{1m} b_{1n} \frac{db_{22}}{dy} - 2b_{1m} b_{2n} \frac{db_{12}}{dy} + 4b_{3m} b_{3n} \frac{db_{33}}{dy} - b_{im} b_{in} \frac{db_{33}}{dy} \right) = \end{aligned} \tag{A.45}$$

Appendix A.6. imjnmkn

$$-\overline{u_i u_j u_k} = C_6 \frac{k^3}{\epsilon} \left(b_{im} b_{jn} \frac{\partial b_{kn}}{\partial x_m} + b_{jm} b_{kn} \frac{\partial b_{in}}{\partial x_m} + b_{km} b_{in} \frac{\partial b_{jn}}{\partial x_m} \right) \tag{A.46}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_6 \frac{k^3}{\epsilon} \left(b_{12} b_{2n} \frac{db_{2n}}{dy} + b_{22} b_{2n} \frac{db_{1n}}{dy} + b_{22} b_{1n} \frac{db_{2n}}{dy} \right) + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.47}$$

At the centreline of channel flow

$$D_{122}(0) = C_6 \frac{k^3}{\epsilon} \left(b_{22} b_{22} + b_{22} b_{11} \right) \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.48}$$

$$D_{112} = -\overline{uuv} = C_6 \frac{k^3}{\epsilon} \left(b_{12} b_{1n} \frac{db_{2n}}{dy} + b_{12} b_{2n} \frac{db_{1n}}{dy} + b_{22} b_{1n} \frac{db_{1n}}{dy} \right) \tag{A.49}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_6 \frac{k^3}{\epsilon} b_{22} b_{2n} \frac{db_{2n}}{dy} + 2 \left(C_6 \frac{k^3}{\epsilon} b_{22} b_{2n} \frac{db_{2n}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.50}$$

$$D_{332} = -\overline{wwv} = C_6 \frac{k^3}{\epsilon} b_{22} b_{33} \frac{db_{33}}{dy} \tag{A.51}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.52}$$

$$b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \qquad () \tag{A.53}$$

$$D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \qquad () \tag{A.54}$$

Appendix A.7. imjnnkm

$$-\overline{u_i u_j u_k} = C_7 \frac{k^3}{\epsilon} \left(b_{im} b_{jn} \frac{\partial b_{km}}{\partial x_n} + b_{jm} b_{kn} \frac{\partial b_{im}}{\partial x_n} + b_{km} b_{in} \frac{\partial b_{jm}}{\partial x_n} \right) \tag{A.55}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_7 \frac{k^3}{\epsilon} \left(b_{1m} b_{22} \frac{db_{2m}}{dy} + b_{2m} b_{22} \frac{db_{1m}}{dy} + b_{2m} b_{12} \frac{db_{2m}}{dy} \right) + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.56}$$

At the centreline of channel flow

$$D_{122}(0) = C_7 \frac{k^3}{\epsilon} \left(b_{11} b_{22} + b_{22} b_{22} \right) \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.57}$$

$$D_{112} = -\overline{uuv} = C_7 \frac{k^3}{\epsilon} \left(b_{1m} b_{12} \frac{db_{2m}}{dy} + b_{1m} b_{22} \frac{db_{1m}}{dy} + b_{2m} b_{12} \frac{db_{1m}}{dy} \right) \tag{A.58}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_7 \frac{k^3}{\epsilon} b_{2m} b_{22} \frac{db_{2m}}{dy} + 2 \left(C_7 \frac{k^3}{\epsilon} b_{2m} b_{22} \frac{db_{2m}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.59}$$

$$D_{332} = -\overline{wwv} = C_7 \frac{k^3}{\epsilon} b_{33} b_{22} \frac{db_{33}}{dy} \tag{A.60}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.61}$$

$$b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \tag{A.62}$$

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$$D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \tag{A.63}$$

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Appendix A.8. immnjkn

$$-\overline{u_i u_j u_k} = C_8 \frac{k^3}{\epsilon} \left(b_{im} b_{mn} \frac{\partial b_{kn}}{\partial x_j} + b_{jm} b_{mn} \frac{\partial b_{in}}{\partial x_k} + b_{km} b_{mn} \frac{\partial b_{jn}}{\partial x_i} \right) \tag{A.64}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_8 \frac{k^3}{\epsilon} \left(b_{1m} b_{mn} \frac{db_{2n}}{dy} + b_{2m} b_{mn} \frac{db_{1n}}{dy} \right) + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.65}$$

At the centreline of channel flow

$$D_{122}(0) = C_8 \frac{k^3}{\epsilon} \left(b_{11} b_{11} + b_{22} b_{22} \right) \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up} \right) \tag{A.66}$$

$$D_{112} = -\overline{uuv} = C_8 \frac{k^3}{\epsilon} b_{1m} b_{mn} \frac{db_{1n}}{dy} \tag{A.67}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_8 \frac{k^3}{\epsilon} b_{2m} b_{mn} \frac{db_{2n}}{dy} + 2 \left(C_8 \frac{k^3}{\epsilon} b_{2m} b_{mn} \frac{db_{2n}}{dy} - \frac{1}{\rho}\overline{vp} \right) \tag{A.68}$$

$$D_{332} = -\overline{wwv} = C_8 \frac{k^3}{\epsilon} b_{33} b_{33} \frac{db_{33}}{dy} \tag{A.69}$$

$$\frac{1}{\rho}\overline{u_i p} = C_8 \frac{k^3}{\epsilon} b_{im} b_{mn} \frac{\partial b_{jn}}{\partial x_j} \text{ or } C_8 \frac{k^3}{\epsilon} b_{jm} b_{mn} \frac{\partial b_{in}}{\partial x_j} \tag{A.70}$$

$$\begin{aligned} b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \\ C_8 \frac{k^3}{\epsilon} b_{mn} \left(b_{11} b_{2m} \frac{db_{2n}}{dy} + b_{22} b_{1m} \frac{db_{1n}}{dy} - 2 b_{12} b_{1m} \frac{db_{2n}}{dy} + 4 b_{33} b_{3m} \frac{db_{3n}}{dy} - b_{33} b_{im} \frac{db_{in}}{dy} \right) = \end{aligned} \tag{A.71}$$

$$\begin{aligned} D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ C_8 \frac{k^3}{\epsilon} b_{mn} \left(b_{2m} \frac{db_{2n}}{dy} \frac{db_{11}}{dy} + b_{1m} \frac{db_{1n}}{dy} \frac{db_{22}}{dy} - 2 b_{1m} \frac{db_{2n}}{dy} \frac{db_{12}}{dy} + 4 b_{3m} \frac{db_{3n}}{dy} \frac{db_{33}}{dy} - b_{im} \frac{db_{in}}{dy} \frac{db_{33}}{dy} \right) = \end{aligned} \tag{A.72}$$

or

$$\begin{aligned} b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \\ C_8 \frac{k^3}{\epsilon} b_{mn} \left(b_{11} b_{2m} \frac{db_{2n}}{dy} + b_{22} b_{1m} \frac{db_{1n}}{dy} - 2 b_{12} b_{2m} \frac{db_{1n}}{dy} + 4 b_{33} b_{3m} \frac{db_{3n}}{dy} - b_{33} b_{im} \frac{db_{in}}{dy} \right) = \end{aligned} \tag{A.73}$$

$$\begin{aligned} D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ C_8 \frac{k^3}{\epsilon} b_{mn} \left(b_{2m} \frac{db_{2n}}{dy} \frac{db_{11}}{dy} + b_{1m} \frac{db_{1n}}{dy} \frac{db_{22}}{dy} - 2 b_{2m} \frac{db_{1n}}{dy} \frac{db_{12}}{dy} + 4 b_{3m} \frac{db_{3n}}{dy} \frac{db_{33}}{dy} - b_{im} \frac{db_{in}}{dy} \frac{db_{33}}{dy} \right) = \end{aligned} \tag{A.74}$$

Appendix A.9. immnkjn

$$-\overline{u_i u_j u_k} = C_9 \frac{k^3}{\epsilon}\left(b_{im} b_{mn} \frac{\partial b_{jn}}{\partial x_k} + b_{jm} b_{mn} \frac{\partial b_{kn}}{\partial x_i} + b_{km} b_{mn} \frac{\partial b_{in}}{\partial x_j}\right) \tag{A.75}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_9 \frac{k^3}{\epsilon}\left(b_{1m} b_{mn} \frac{db_{2n}}{dy} + b_{2m} b_{mn} \frac{db_{1n}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.76}$$

At the centreline of channel flow

$$D_{122}(0) = C_9 \frac{k^3}{\epsilon}\left(b_{11} b_{11} + b_{22} b_{22}\right) \frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.77}$$

$$D_{112} = -\overline{uuv} = C_9 \frac{k^3}{\epsilon} b_{1m} b_{mn} \frac{db_{1n}}{dy} \tag{A.78}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_9 \frac{k^3}{\epsilon} b_{2m} b_{mn} \frac{db_{2n}}{dy} + 2\left(C_9 \frac{k^3}{\epsilon} b_{2m} b_{mn} \frac{db_{2n}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.79}$$

$$D_{332} = -\overline{wwv} = C_9 \frac{k^3}{\epsilon} b_{33} b_{33} \frac{db_{33}}{dy} \tag{A.80}$$

$$\frac{1}{\rho}\overline{u_i p} = C_9 \frac{k^3}{\epsilon} b_{jm} b_{mn} \frac{\partial b_{in}}{\partial x_j} \text{ or } C_9 \frac{k^3}{\epsilon} b_{im} b_{mn} \frac{\partial b_{jn}}{\partial x_j} \tag{A.81}$$

$$\begin{aligned} &b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \\ &\qquad C_9 \frac{k^3}{\epsilon} b_{mn}\left(b_{11} b_{2m} \frac{db_{2n}}{dy} + b_{22} b_{1m} \frac{db_{1n}}{dy} - 2 b_{12} b_{1m} \frac{db_{2n}}{dy} + 4 b_{33} b_{3m} \frac{db_{3n}}{dy} - b_{33} b_{im} \frac{db_{in}}{dy}\right) = \end{aligned} \tag{A.82}$$

$$\begin{aligned} &D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ &\qquad C_9 \frac{k^3}{\epsilon} b_{mn}\left(b_{2m} \frac{db_{2n}}{dy} \frac{db_{11}}{dy} + b_{1m} \frac{db_{1n}}{dy} \frac{db_{22}}{dy} - 2 b_{1m} \frac{db_{2n}}{dy} \frac{db_{12}}{dy} + 4 b_{3m} \frac{db_{3n}}{dy} \frac{db_{33}}{dy} - b_{im} \frac{db_{in}}{dy} \frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.83}$$

or

$$\begin{aligned} &b_{11} D_{222} + b_{22} D_{112} - 2 b_{12} D_{122} + 4 b_{33} D_{332} - b_{33} D_{ii2} = \\ &\qquad C_9 \frac{k^3}{\epsilon} b_{mn}\left(b_{11} b_{2m} \frac{db_{2n}}{dy} + b_{22} b_{1m} \frac{db_{1n}}{dy} - 2 b_{12} b_{2m} \frac{db_{1n}}{dy} + 4 b_{33} b_{3m} \frac{db_{3n}}{dy} - b_{33} b_{im} \frac{db_{in}}{dy}\right) = \end{aligned} \tag{A.84}$$

$$\begin{aligned} &D_{222} \frac{db_{11}}{dy} + D_{112} \frac{db_{22}}{dy} - 2 D_{122} \frac{db_{12}}{dy} + 4 D_{332} \frac{db_{33}}{dy} - D_{ii2} \frac{db_{33}}{dy} = \\ &\qquad C_9 \frac{k^3}{\epsilon} b_{mn}\left(b_{2m} \frac{db_{2n}}{dy} \frac{db_{11}}{dy} + b_{1m} \frac{db_{1n}}{dy} \frac{db_{22}}{dy} - 2 b_{2m} \frac{db_{1n}}{dy} \frac{db_{12}}{dy} + 4 b_{3m} \frac{db_{3n}}{dy} \frac{db_{33}}{dy} - b_{im} \frac{db_{in}}{dy} \frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.85}$$

Appendix A.10. immnnjk

$$-\overline{u_i u_j u_k} = C_{10}\frac{k^3}{\epsilon}\left(b_{im}b_{mn}\frac{\partial b_{jk}}{\partial x_n} + b_{jm}b_{mn}\frac{\partial b_{ki}}{\partial x_n} + b_{km}b_{mn}\frac{\partial b_{ij}}{\partial x_n}\right) \tag{A.86}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{10}\frac{k^3}{\epsilon}\left(b_{1m}b_{m2}\frac{db_{22}}{dy} + 2b_{2m}b_{m2}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.87}$$

At the centreline of channel flow

$$D_{122}(0) = 2C_{10}\frac{k^3}{\epsilon}b_{22}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.88}$$

$$D_{112} = -\overline{uuv} = C_{10}\frac{k^3}{\epsilon}\left(2b_{1m}b_{m2}\frac{db_{12}}{dy} + b_{2m}b_{m2}\frac{db_{11}}{dy}\right) \tag{A.89}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{10}\frac{k^3}{\epsilon}b_{2m}b_{m2}\frac{db_{22}}{dy} + 2\left(C_{10}\frac{k^3}{\epsilon}b_{2m}b_{m2}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.90}$$

$$D_{332} = -\overline{wwv} = C_{10}\frac{k^3}{\epsilon}b_{2m}b_{m2}\frac{db_{33}}{dy} \tag{A.91}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.92}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.93}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.94}$$

Appendix A.11. mnmnijk

$$-\overline{u_i u_j u_k} = C_{11}\frac{k^3}{\epsilon}\left(b_{mn}b_{mn}\frac{\partial b_{jk}}{\partial x_i} + b_{mn}b_{mn}\frac{\partial b_{ki}}{\partial x_j} + b_{mn}b_{mn}\frac{\partial b_{ij}}{\partial x_k}\right) \tag{A.95}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{12}}{dy} + \left(C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.96}$$

At the centreline of channel flow

$$D_{122}(0) = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{12}}{dy} + \left(C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.97}$$

$$D_{112} = -\overline{uuv} = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{11}}{dy} \tag{A.98}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{22}}{dy} + 2\left(C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.99}$$

$$D_{332} = -\overline{wwv} = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{db_{33}}{dy} \tag{A.100}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{\partial b_{ij}}{\partial x_j} \tag{A.101}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\left(b_{11}\frac{db_{22}}{dy} + b_{22}\frac{db_{11}}{dy} - 2b_{12}\frac{db_{12}}{dy} + 4b_{33}\frac{db_{33}}{dy} - b_{33}\times 0\right) = C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\frac{d}{dy}\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) = 0 \end{aligned} \tag{A.102}$$

$$\begin{aligned} D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\left(\frac{db_{22}}{dy}\frac{db_{11}}{dy} + \frac{db_{11}}{dy}\frac{db_{22}}{dy} - 2\frac{db_{12}}{dy}\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy}\frac{db_{33}}{dy} - 0\times\frac{db_{33}}{dy}\right) = \\ 2C_{11}\frac{k^3}{\epsilon}b_{mn}b_{mn}\left(\frac{db_{11}}{dy}\frac{db_{22}}{dy} - \frac{db_{12}}{dy}\frac{db_{12}}{dy} + 2\frac{db_{33}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.103}$$

The above is zero for axis aligned axisymmetric turbulence.

Appendix A.12. ijkmnmn

$$-\overline{u_i u_j u_k} = C_{12}\frac{k^3}{\epsilon}\left(\delta_{ij} b_{km}\frac{\partial b_{mn}}{\partial x_n} + \delta_{jk} b_{im}\frac{\partial b_{mn}}{\partial x_n} + \delta_{ki} b_{jm}\frac{\partial b_{mn}}{\partial x_n}\right) \tag{A.104}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(C_{12}\frac{k^3}{\epsilon} b_{1m}\frac{db_{m2}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.105}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(C_{12}\frac{k^3}{\epsilon} b_{11}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.106}$$

$$D_{112} = -\overline{uuv} = C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy} \tag{A.107}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy} + 2\left(C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.108}$$

$$D_{332} = -\overline{wwv} = C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy} \tag{A.109}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{12}\frac{k^3}{\epsilon} b_{im}\frac{\partial b_{mn}}{\partial x_n} \tag{A.110}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.111}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{12}\frac{k^3}{\epsilon} b_{2m}\frac{db_{m2}}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.112}$$

Appendix A.13. ijmnkmn

$$-\overline{u_i u_j u_k} = C_{13}\frac{k^3}{\epsilon}\left(\delta_{ij} b_{mn}\frac{\partial b_{mn}}{\partial x_k} + \delta_{jk} b_{mn}\frac{\partial b_{mn}}{\partial x_i} + \delta_{ki} b_{mn}\frac{\partial b_{mn}}{\partial x_j}\right) \tag{A.113}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.114}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.115}$$

$$D_{112} = -\overline{uuv} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy} \tag{A.116}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy} + 2\left(C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.117}$$

$$D_{332} = -\overline{wwv} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy} \tag{A.118}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{\partial b_{mn}}{\partial x_i} \tag{A.119}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.120}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{13}\frac{k^3}{\epsilon} b_{mn}\frac{db_{mn}}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.121}$$

Appendix A.14. ijmnmkn

$$-\overline{u_i u_j u_k} = C_{14}\frac{k^3}{\epsilon}\left(\delta_{ij} b_{mn}\frac{\partial b_{kn}}{\partial x_m} + \delta_{jk} b_{mn}\frac{\partial b_{in}}{\partial x_m} + \delta_{ki} b_{mn}\frac{\partial b_{jn}}{\partial x_m}\right) \tag{A.122}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{1n}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.123}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(C_{14}\frac{k^3}{\epsilon} b_{22}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.124}$$

$$D_{112} = -\overline{uuv} = C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy} \tag{A.125}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy} + 2\left(C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.126}$$

$$D_{332} = -\overline{wwv} = C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy} \tag{A.127}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{14}\frac{k^3}{\epsilon} b_{mn}\frac{\partial b_{in}}{\partial x_m} \tag{A.128}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.129}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{14}\frac{k^3}{\epsilon} b_{2n}\frac{db_{2n}}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.130}$$

Appendix A.15. imjmnkn

$$-\overline{u_i u_j u_k} = C_{15}\frac{k^3}{\epsilon}\left(b_{ji}\frac{\partial b_{kn}}{\partial x_n} + b_{kj}\frac{\partial b_{in}}{\partial x_n} + b_{ik}\frac{\partial b_{jn}}{\partial x_n}\right) \tag{A.131}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{15}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{22}}{dy} + b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.132}$$

At the centreline of channel flow

$$D_{122}(0) = C_{15}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.133}$$

$$D_{112} = -\overline{uuv} = C_{15}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + 2b_{12}\frac{db_{12}}{dy}\right) \tag{A.134}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{15}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{15}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.135}$$

$$D_{332} = -\overline{wwv} = C_{15}\frac{k^3}{\epsilon}b_{33}\frac{db_{22}}{dy} \tag{A.136}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.137}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.138}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.139}$$

Appendix A.16. imjnnkm

$$-\overline{u_i u_j u_k} = C_{16}\frac{k^3}{\epsilon}\left(b_{jn}\frac{\partial b_{ki}}{\partial x_n} + b_{kn}\frac{\partial b_{ij}}{\partial x_n} + b_{in}\frac{\partial b_{jk}}{\partial x_n}\right) \tag{A.140}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{16}\frac{k^3}{\epsilon}\left(2b_{22}\frac{db_{12}}{dy} + b_{12}\frac{db_{22}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.141}$$

At the centreline of channel flow

$$D_{122}(0) = 2C_{16}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.142}$$

$$D_{112} = -\overline{uuv} = C_{16}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{12}}{dy} + b_{22}\frac{db_{11}}{dy}\right) \tag{A.143}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{16}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{16}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.144}$$

$$D_{332} = -\overline{wwv} = C_{16}\frac{k^3}{\epsilon}b_{22}\frac{db_{33}}{dy} \tag{A.145}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.146}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \tag{A.147}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \tag{A.148}$$

Appendix A.17. immnnjk

$$-\overline{u_i u_j u_k} = C_{17}\frac{k^3}{\epsilon}\left(b_{in}\frac{\partial b_{jk}}{\partial x_n} + b_{jn}\frac{\partial b_{ki}}{\partial x_n} + b_{kn}\frac{\partial b_{ij}}{\partial x_n}\right) \tag{A.149}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{17}\frac{k^3}{\epsilon}\left(b_{12}\frac{db_{22}}{dy} + 2b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.150}$$

At the centreline of channel flow

$$D_{122}(0) = 2C_{17}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.151}$$

$$D_{112} = -\overline{uuv} = C_{17}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{12}}{dy} + b_{22}\frac{db_{11}}{dy}\right) \tag{A.152}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{17}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{17}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.153}$$

$$D_{332} = -\overline{wwv} = C_{17}\frac{k^3}{\epsilon}b_{22}\frac{db_{33}}{dy} \tag{A.154}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.155}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.156}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.157}$$

Appendix A.18. ijkmnmn

$$-\overline{u_i u_j u_k} = C_{18}\frac{k^3}{\epsilon}\left(b_{ij}\frac{\partial b_{kn}}{\partial x_n} + b_{jk}\frac{\partial b_{in}}{\partial x_n} + b_{ki}\frac{\partial b_{jn}}{\partial x_n}\right) \tag{A.158}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{18}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{22}}{dy} + b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.159}$$

At the centreline of channel flow

$$D_{122}(0) = C_{18}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.160}$$

$$D_{112} = -\overline{uuv} = C_{18}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + 2b_{12}\frac{db_{12}}{dy}\right) \tag{A.161}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{18}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{18}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.162}$$

$$D_{332} = -\overline{wwv} = C_{18}\frac{k^3}{\epsilon}b_{33}\frac{db_{22}}{dy} \tag{A.163}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.164}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \tag{A.165}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \tag{A.166}$$

Appendix A.19. ijmmnkn

$$-\overline{u_i u_j u_k} = C_{19}\frac{k^3}{\epsilon}\left(b_{ij}\frac{\partial b_{kn}}{\partial x_n} + b_{jk}\frac{\partial b_{in}}{\partial x_n} + b_{ki}\frac{\partial b_{jn}}{\partial x_n}\right) \tag{A.167}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{19}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{22}}{dy} + b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.168}$$

At the centreline of channel flow

$$D_{122}(0) = C_{19}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.169}$$

$$D_{112} = -\overline{uuv} = C_{19}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + 2b_{12}\frac{db_{12}}{dy}\right) \tag{A.170}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{19}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{19}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.171}$$

$$D_{332} = -\overline{wwv} = C_{19}\frac{k^3}{\epsilon}b_{33}\frac{db_{22}}{dy} \tag{A.172}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.173}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.174}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.175}$$

Appendix A.20. ijmnmkn

$$-\overline{u_i u_j u_k} = C_{20}\frac{k^3}{\epsilon}\left(b_{ij}\frac{\partial b_{km}}{\partial x_m} + b_{jk}\frac{\partial b_{im}}{\partial x_m} + b_{ki}\frac{\partial b_{jm}}{\partial x_m}\right) \tag{A.176}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{20}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{22}}{dy} + b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.177}$$

At the centreline of channel flow

$$D_{122}(0) = C_{20}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.178}$$

$$D_{112} = -\overline{uuv} = C_{20}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + 2b_{12}\frac{db_{12}}{dy}\right) \tag{A.179}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{20}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{20}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.180}$$

$$D_{332} = -\overline{wwv} = C_{20}\frac{k^3}{\epsilon}b_{33}\frac{db_{22}}{dy} \tag{A.181}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.182}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \tag{A.183}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \tag{A.184}$$

Appendix A.21. imjmnkn

$$-\overline{u_i u_j u_k} = C_{21}\frac{k^3}{\epsilon}\left(b_{ij}\frac{\partial b_{kn}}{\partial x_n} + b_{jk}\frac{\partial b_{in}}{\partial x_n} + b_{ki}\frac{\partial b_{jn}}{\partial x_n}\right) \tag{A.185}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{21}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{22}}{dy} + b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.186}$$

At the centreline of channel flow

$$D_{122}(0) = C_{21}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.187}$$

$$D_{112} = -\overline{uuv} = C_{21}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + 2b_{12}\frac{db_{12}}{dy}\right) \tag{A.188}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{21}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{21}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.189}$$

$$D_{332} = -\overline{wwv} = C_{21}\frac{k^3}{\epsilon}b_{33}\frac{db_{22}}{dy} \tag{A.190}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.191}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \tag{A.192}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \tag{A.193}$$

Appendix A.22. imnnjkm

$$-\overline{u_i u_j u_k} = C_{22}\frac{k^3}{\epsilon}\left(b_{im}\frac{\partial b_{km}}{\partial x_j} + b_{jm}\frac{\partial b_{im}}{\partial x_k} + b_{km}\frac{\partial b_{jm}}{\partial x_i}\right) \tag{A.194}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{22}\frac{k^3}{\epsilon}\left(b_{1m}\frac{db_{2m}}{dy} + b_{2m}\frac{db_{1m}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.195}$$

At the centreline of channel flow

$$D_{122}(0) = C_{22}\frac{k^3}{\epsilon}\left(b_{11} + b_{22}\right)\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.196}$$

$$D_{112} = -\overline{uuv} = C_{22}\frac{k^3}{\epsilon}b_{1m}\frac{db_{1m}}{dy} \tag{A.197}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{22}\frac{k^3}{\epsilon}b_{2m}\frac{db_{2m}}{dy} + 2\left(C_{22}\frac{k^3}{\epsilon}b_{2m}\frac{db_{2m}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.198}$$

$$D_{332} = -\overline{wwv} = C_{22}\frac{k^3}{\epsilon}b_{33}\frac{db_{33}}{dy} \tag{A.199}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{22}\frac{k^3}{\epsilon}b_{im}\frac{\partial b_{jm}}{\partial x_j} \text{ or } C_{22}\frac{k^3}{\epsilon}b_{jm}\frac{\partial b_{im}}{\partial x_j} \tag{A.200}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\qquad C_{22}\frac{k^3}{\epsilon}\left(b_{11}b_{2m}\frac{db_{2m}}{dy} + b_{22}b_{1m}\frac{db_{1m}}{dy} - 2b_{12}b_{1m}\frac{db_{2m}}{dy} + 4b_{33}b_{3m}\frac{db_{3m}}{dy} - b_{33}b_{im}\frac{db_{im}}{dy}\right) = \end{aligned} \tag{A.201}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\qquad C_{22}\frac{k^3}{\epsilon}\left(b_{2m}\frac{db_{2m}}{dy}\frac{db_{11}}{dy} + b_{1m}\frac{db_{1m}}{dy}\frac{db_{22}}{dy} - 2b_{1m}\frac{db_{2m}}{dy}\frac{db_{12}}{dy} + 4b_{3m}\frac{db_{3m}}{dy}\frac{db_{33}}{dy} - b_{im}\frac{db_{im}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.202}$$

or

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\qquad C_{22}\frac{k^3}{\epsilon}\left(b_{11}b_{2m}\frac{db_{2m}}{dy} + b_{22}b_{1m}\frac{db_{1m}}{dy} - 2b_{12}b_{2m}\frac{db_{1m}}{dy} + 4b_{33}b_{3m}\frac{db_{3m}}{dy} - b_{33}b_{im}\frac{db_{im}}{dy}\right) = \end{aligned} \tag{A.203}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\qquad C_{22}\frac{k^3}{\epsilon}\left(b_{2m}\frac{db_{2m}}{dy}\frac{db_{11}}{dy} + b_{1m}\frac{db_{1m}}{dy}\frac{db_{22}}{dy} - 2b_{2m}\frac{db_{1m}}{dy}\frac{db_{12}}{dy} + 4b_{3m}\frac{db_{3m}}{dy}\frac{db_{33}}{dy} - b_{im}\frac{db_{im}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.204}$$

Appendix A.23. imnnkjm

$$-\overline{u_i u_j u_k} = C_{23}\frac{k^3}{\epsilon}\left(b_{im}\frac{\partial b_{jm}}{\partial x_k} + b_{jm}\frac{\partial b_{km}}{\partial x_i} + b_{km}\frac{\partial b_{im}}{\partial x_j}\right) \tag{A.205}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{23}\frac{k^3}{\epsilon}\left(b_{1m}\frac{db_{2m}}{dy} + b_{2m}\frac{db_{1m}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.206}$$

At the centreline of channel flow

$$D_{122}(0) = C_{23}\frac{k^3}{\epsilon}\left(b_{11} + b_{22}\right)\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.207}$$

$$D_{112} = -\overline{uuv} = C_{23}\frac{k^3}{\epsilon}b_{1m}\frac{db_{1m}}{dy} \tag{A.208}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{23}\frac{k^3}{\epsilon}b_{2m}\frac{db_{2m}}{dy} + 2\left(C_{23}\frac{k^3}{\epsilon}b_{2m}\frac{db_{2m}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.209}$$

$$D_{332} = -\overline{wwv} = C_{23}\frac{k^3}{\epsilon}b_{33}\frac{db_{33}}{dy} \tag{A.210}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{23}\frac{k^3}{\epsilon}b_{im}\frac{\partial b_{jm}}{\partial x_j} \text{ or } C_{23}\frac{k^3}{\epsilon}b_{jm}\frac{\partial b_{im}}{\partial x_j} \tag{A.211}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\qquad C_{23}\frac{k^3}{\epsilon}\left(b_{11}b_{2m}\frac{db_{2m}}{dy} + b_{22}b_{1m}\frac{db_{1m}}{dy} - 2b_{12}b_{1m}\frac{db_{2m}}{dy} + 4b_{33}b_{3m}\frac{db_{3m}}{dy} - b_{33}b_{im}\frac{db_{im}}{dy}\right) = \end{aligned} \tag{A.212}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\qquad C_{23}\frac{k^3}{\epsilon}\left(b_{2m}\frac{db_{2m}}{dy}\frac{db_{11}}{dy} + b_{1m}\frac{db_{1m}}{dy}\frac{db_{22}}{dy} - 2b_{1m}\frac{db_{2m}}{dy}\frac{db_{12}}{dy} + 4b_{3m}\frac{db_{3m}}{dy}\frac{db_{33}}{dy} - b_{im}\frac{db_{im}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.213}$$

or

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\qquad C_{23}\frac{k^3}{\epsilon}\left(b_{11}b_{2m}\frac{db_{2m}}{dy} + b_{22}b_{1m}\frac{db_{1m}}{dy} - 2b_{12}b_{2m}\frac{db_{1m}}{dy} + 4b_{33}b_{3m}\frac{db_{3m}}{dy} - b_{33}b_{im}\frac{db_{im}}{dy}\right) = \end{aligned} \tag{A.214}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\qquad C_{23}\frac{k^3}{\epsilon}\left(b_{2m}\frac{db_{2m}}{dy}\frac{db_{11}}{dy} + b_{1m}\frac{db_{1m}}{dy}\frac{db_{22}}{dy} - 2b_{2m}\frac{db_{1m}}{dy}\frac{db_{12}}{dy} + 4b_{3m}\frac{db_{3m}}{dy}\frac{db_{33}}{dy} - b_{im}\frac{db_{im}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.215}$$

Appendix A.24. imnnmjk

$$-\overline{u_i u_j u_k} = C_{24}\frac{k^3}{\epsilon}\left(b_{im}\frac{\partial b_{jk}}{\partial x_m} + b_{jm}\frac{\partial b_{ki}}{\partial x_m} + b_{km}\frac{\partial b_{ij}}{\partial x_m}\right) \tag{A.216}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{24}\frac{k^3}{\epsilon}\left(b_{12}\frac{db_{22}}{dy} + 2b_{22}\frac{db_{12}}{dy}\right) + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.217}$$

At the centreline of channel flow

$$D_{122}(0) = 2C_{24}\frac{k^3}{\epsilon}b_{22}\frac{db_{12}}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.218}$$

$$D_{112} = -\overline{uuv} = C_{24}\frac{k^3}{\epsilon}\left(2b_{12}\frac{db_{12}}{dy} + b_{22}\frac{db_{11}}{dy}\right) \tag{A.219}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{24}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} + 2\left(C_{24}\frac{k^3}{\epsilon}b_{22}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.220}$$

$$D_{332} = -\overline{wwv} = C_{24}\frac{k^3}{\epsilon}b_{22}\frac{db_{33}}{dy} \tag{A.221}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.222}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.223}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.224}$$

Appendix A.25. ijkmmnn

$$-\overline{u_i u_j u_k} = C_{25}\frac{k^2}{\epsilon}\left(b_{ij}b_{km}\frac{\partial k}{\partial x_m} + b_{jk}b_{im}\frac{\partial k}{\partial x_m} + b_{ki}b_{jm}\frac{\partial k}{\partial x_m}\right) \tag{A.225}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 3C_{25}\frac{k^2}{\epsilon}b_{12}b_{22}\frac{dk}{dy} + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.226}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.227}$$

$$D_{112} = -\overline{uuv} = C_{25}\frac{k^2}{\epsilon}\left(b_{11}b_{22}\frac{dk}{dy} + 2b_{12}b_{12}\frac{dk}{dy}\right) \tag{A.228}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{25}\frac{k^2}{\epsilon}b_{22}b_{22}\frac{dk}{dy} + 2\left(C_{25}\frac{k^2}{\epsilon}b_{22}b_{22}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.229}$$

$$D_{332} = -\overline{wwv} = C_{25}\frac{k^2}{\epsilon}b_{33}b_{22}\frac{dk}{dy} \tag{A.230}$$

$$\frac{1}{\rho}\overline{u_i p} = \tag{A.231}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \qquad () \tag{A.232}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \qquad () \tag{A.233}$$

Appendix A.26. imjmknn

$$-\overline{u_i u_j u_k} = C_{26}\frac{k^2}{\epsilon}\left(b_{im}b_{jm}\frac{\partial k}{\partial x_k} + b_{jm}b_{km}\frac{\partial k}{\partial x_i} + b_{km}b_{im}\frac{\partial k}{\partial x_j}\right) \tag{A.234}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{26}\frac{k^2}{\epsilon}b_{1m}b_{2m}\frac{dk}{dy} + \left(C_{26}\frac{k^2}{\epsilon}b_{1m}b_{2m}\frac{dk}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.235}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.236}$$

$$D_{112} = -\overline{uuv} = C_{26}\frac{k^2}{\epsilon}b_{1m}b_{1m}\frac{dk}{dy} \tag{A.237}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{26}\frac{k^2}{\epsilon}b_{2m}b_{2m}\frac{dk}{dy} + 2\left(C_{26}\frac{k^2}{\epsilon}b_{2m}b_{2m}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.238}$$

$$D_{332} = -\overline{wwv} = C_{26}\frac{k^2}{\epsilon}b_{33}b_{33}\frac{dk}{dy} \tag{A.239}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{26}\frac{k^2}{\epsilon}b_{im}b_{jm}\frac{\partial k}{\partial x_j} \tag{A.240}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &C_{26}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{11}b_{2m}b_{2m} + b_{22}b_{1m}b_{1m} - 2b_{12}b_{1m}b_{2m} + 4b_{33}b_{33}b_{33} - b_{33}b_{ij}b_{ij}\right) = C_{26}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(6b_{33}^3 - b_{33}b_{ij}b_{ij}\right) \end{aligned} \tag{A.241}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\qquad C_{26}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{2m}b_{2m}\frac{db_{11}}{dy} + b_{1m}b_{1m}\frac{db_{22}}{dy} - 2b_{1m}b_{2m}\frac{db_{12}}{dy} + 4b_{33}b_{33}\frac{db_{33}}{dy} - b_{ij}b_{ij}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.242}$$

Appendix A.27. immnnjk

$$-\overline{u_i u_j u_k} = C_{27}\frac{k^2}{\epsilon}\left(\delta_{jk}b_{im}b_{mn}\frac{\partial k}{\partial x_n} + \delta_{ki}b_{jm}b_{mn}\frac{\partial k}{\partial x_n} + \delta_{ij}b_{km}b_{mn}\frac{\partial k}{\partial x_n}\right) \tag{A.243}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(C_{27}\frac{k^2}{\epsilon}b_{1m}b_{m2}\frac{dk}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.244}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.245}$$

$$D_{112} = -\overline{uuv} = C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy} \tag{A.246}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy} + 2\left(C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.247}$$

$$D_{332} = -\overline{wwv} = C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy} \tag{A.248}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{27}\frac{k^2}{\epsilon}b_{im}b_{mn}\frac{\partial k}{\partial x_n} \tag{A.249}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.250}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{27}\frac{k^2}{\epsilon}b_{2m}b_{m2}\frac{dk}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.251}$$

Appendix A.28. mnmnijk

$$-\overline{u_i u_j u_k} = C_{28}\frac{k^2}{\epsilon}\left(\delta_{jk}b_{mn}b_{mn}\frac{\partial k}{\partial x_i} + \delta_{ki}b_{mn}b_{mn}\frac{\partial k}{\partial x_j} + \delta_{ij}b_{mn}b_{mn}\frac{\partial k}{\partial x_k}\right) \tag{A.252}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.253}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.254}$$

$$D_{112} = -\overline{uuv} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy} \tag{A.255}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy} + 2\left(C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.256}$$

$$D_{332} = -\overline{wwv} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy} \tag{A.257}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{\partial k}{\partial x_i} \tag{A.258}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.259}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{28}\frac{k^2}{\epsilon}b_{mn}b_{mn}\frac{dk}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.260}$$

Appendix A.29. ijkmnmn

$$-\overline{u_i u_j u_k} = C_{29}\frac{k^3}{\epsilon}\left(\delta_{ij}\frac{\partial b_{kn}}{\partial x_n} + \delta_{jk}\frac{\partial b_{in}}{\partial x_n} + \delta_{ki}\frac{\partial b_{jn}}{\partial x_n}\right) \tag{A.261}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(C_{29}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.262}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(C_{29}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.263}$$

$$D_{112} = -\overline{uuv} = C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} \tag{A.264}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} + 2\left(C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.265}$$

$$D_{332} = -\overline{wwv} = C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} \tag{A.266}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{29}\frac{k^3}{\epsilon}\frac{\partial b_{in}}{\partial x_n} \tag{A.267}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.268}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{29}\frac{k^3}{\epsilon}\frac{db_{22}}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.269}$$

Appendix A.30. imjnkmn

$$-\overline{u_i u_j u_k} = C_{30}\frac{k^3}{\epsilon}\left(\frac{\partial b_{ij}}{\partial x_k} + \frac{\partial b_{jk}}{\partial x_i} + \frac{\partial b_{ki}}{\partial x_j}\right) \tag{A.270}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{30}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} + \left(C_{30}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.271}$$

At the centreline of channel flow

$$D_{122}(0) = C_{30}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} + \left(C_{30}\frac{k^3}{\epsilon}\frac{db_{12}}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.272}$$

$$D_{112} = -\overline{uuv} = C_{30}\frac{k^3}{\epsilon}\frac{db_{11}}{dy} \tag{A.273}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{30}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} + 2\left(C_{30}\frac{k^3}{\epsilon}\frac{db_{22}}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.274}$$

$$D_{332} = -\overline{wwv} = C_{30}\frac{k^3}{\epsilon}\frac{db_{33}}{dy} \tag{A.275}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{30}\frac{k^3}{\epsilon}\frac{\partial b_{ij}}{\partial x_j} \tag{A.276}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\quad C_{30}\frac{k^3}{\epsilon}\left(b_{11}\frac{db_{22}}{dy} + b_{22}\frac{db_{11}}{dy} - 2b_{12}\frac{db_{12}}{dy} + 4b_{33}\frac{db_{33}}{dy} - b_{33}\times 0\right) = C_{30}\frac{k^3}{\epsilon}\frac{d}{dy}\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) = 0 \end{aligned} \tag{A.277}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\quad C_{30}\frac{k^3}{\epsilon}\left(\frac{db_{22}}{dy}\frac{db_{11}}{dy} + \frac{db_{11}}{dy}\frac{db_{22}}{dy} - 2\frac{db_{12}}{dy}\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy}\frac{db_{33}}{dy} - 0\times\frac{db_{33}}{dy}\right) = \\ &\quad 2C_{30}\frac{k^3}{\epsilon}\left(\frac{db_{11}}{dy}\frac{db_{22}}{dy} - \frac{db_{12}}{dy}\frac{db_{12}}{dy} + 2\frac{db_{33}}{dy}\frac{db_{33}}{dy}\right) = \end{aligned} \tag{A.278}$$

The above is zero for axis aligned axisymmetric turbulence.

Appendix A.31. ijkmmnn

$$-\overline{u_i u_j u_k} = C_{31}\frac{k^2}{\epsilon}\left(\delta_{ij}b_{km}\frac{\partial k}{\partial x_m} + \delta_{jk}b_{im}\frac{\partial k}{\partial x_m} + \delta_{ki}b_{jm}\frac{\partial k}{\partial x_m}\right) \tag{A.279}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(C_{31}\frac{k^2}{\epsilon}b_{12}\frac{dk}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.280}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.281}$$

$$D_{112} = -\overline{uuv} = C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} \tag{A.282}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} + 2\left(C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.283}$$

$$D_{332} = -\overline{wwv} = C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} \tag{A.284}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{31}\frac{k^2}{\epsilon}b_{ij}\frac{\partial k}{\partial x_j} \tag{A.285}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - b_{33}\times 3\right) = 0 \tag{A.286}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{31}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\times\frac{db_{33}}{dy}\right) = 0 \tag{A.287}$$

Appendix A.32. imjmknn

$$-\overline{u_i u_j u_k} = C_{32}\frac{k^2}{\epsilon}\left(b_{ji}\frac{\partial k}{\partial x_k} + b_{kj}\frac{\partial k}{\partial x_i} + b_{ik}\frac{\partial k}{\partial x_j}\right) \tag{A.288}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = C_{32}\frac{k^2}{\epsilon}b_{12}\frac{dk}{dy} + \left(C_{32}\frac{k^2}{\epsilon}b_{12}\frac{dk}{dy} - \frac{1}{\rho}\overline{up}\right) \tag{A.289}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.290}$$

$$D_{112} = -\overline{uuv} = C_{32}\frac{k^2}{\epsilon}b_{11}\frac{dk}{dy} \tag{A.291}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{32}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} + 2\left(C_{32}\frac{k^2}{\epsilon}b_{22}\frac{dk}{dy} - \frac{1}{\rho}\overline{vp}\right) \tag{A.292}$$

$$D_{332} = -\overline{wwv} = C_{32}\frac{k^2}{\epsilon}b_{33}\frac{dk}{dy} \tag{A.293}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{32}\frac{k^2}{\epsilon}b_{ij}\frac{\partial k}{\partial x_j} \tag{A.294}$$

$$\begin{aligned} &b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = \\ &\quad C_{32}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{11}b_{22} + b_{22}b_{11} - 2b_{12}b_{12} + 4b_{33}b_{33} - b_{33}\times 0\right) = 2C_{32}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) = 0 \end{aligned} \tag{A.295}$$

$$\begin{aligned} &D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = \\ &\quad C_{32}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{22}\frac{db_{11}}{dy} + b_{11}\frac{db_{22}}{dy} - 2b_{12}\frac{db_{12}}{dy} + 4b_{33}\frac{db_{33}}{dy} - 0\times\frac{db_{33}}{dy}\right) = C_{32}\frac{k^2}{\epsilon}\frac{dk}{dy}\frac{d}{dy}\left(b_{11}b_{22} - b_{12}^2 + 2b_{33}^2\right) = 0 \end{aligned} \tag{A.296}$$

Appendix A.33. ijkmmnn

$$-\overline{u_i u_j u_k} = C_{33}\frac{k^2}{\epsilon}\left(\delta_{ij}\frac{\partial k}{\partial x_k} + \delta_{jk}\frac{\partial k}{\partial x_i} + \delta_{ki}\frac{\partial k}{\partial x_j}\right) \tag{A.297}$$

$$D_{122} = -\overline{uvv} - \frac{1}{\rho}\overline{up} = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.298}$$

At the centreline of channel flow

$$D_{122}(0) = 0 + \left(0 - \frac{1}{\rho}\overline{up}\right) \tag{A.299}$$

$$D_{112} = -\overline{uuv} = C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy} \tag{A.300}$$

$$D_{222} = -\overline{vvv} - \frac{2}{\rho}\overline{vp} = C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy} + 2\left(C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy} + -\frac{1}{\rho}\overline{vp}\right) \tag{A.301}$$

$$D_{332} = -\overline{wwv} = C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy} \tag{A.302}$$

$$\frac{1}{\rho}\overline{u_i p} = C_{33}\frac{k^2}{\epsilon}\frac{\partial k}{\partial x_i} \tag{A.303}$$

$$b_{11}D_{222} + b_{22}D_{112} - 2b_{12}D_{122} + 4b_{33}D_{332} - b_{33}D_{ii2} = C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(b_{11} + b_{22} - 2b_{12}\times 0 + 4b_{33} - 3b_{33}\right) = 0 \tag{A.304}$$

$$D_{222}\frac{db_{11}}{dy} + D_{112}\frac{db_{22}}{dy} - 2D_{122}\frac{db_{12}}{dy} + 4D_{332}\frac{db_{33}}{dy} - D_{ii2}\frac{db_{33}}{dy} = C_{33}\frac{k^2}{\epsilon}\frac{dk}{dy}\left(\frac{db_{11}}{dy} + \frac{db_{22}}{dy} - 0\times\frac{db_{12}}{dy} + 4\frac{db_{33}}{dy} - 3\frac{db_{33}}{dy}\right) = 0 \tag{A.305}$$

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